UNIT-3:

1. $p\tan x + q\tan y = \tan z$

Sol: Given,

$$p\tan x + q\tan y = \tan z$$

It is in the Form of Lagranger linear Equations

$$Pp + Qq = R$$

$\Rightarrow P = \tan x \quad Q = \tan y \quad R = \tan z$

Auxilary Equations $\Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\Rightarrow \frac{dx}{\tan x} = \frac{dy}{\tan y} = \frac{dz}{\tan z}$$

Consider $\frac{dx}{\tan x} = \frac{dy}{\tan y}$

$\Rightarrow \cot x\, dx = \cot y\, dy$

Integrate on both sides

$\Rightarrow \int \cot x\, dx = \int \cot y\, dy$

$\Rightarrow \log|\sin x| = \log|\sin y| + C$

$\Rightarrow \log|\sin x| - \log|\sin y| = C$

$\Rightarrow \log\left(\frac{\sin x}{\sin y}\right) = C$

$\Rightarrow \frac{\sin x}{\sin y} = C_1 = u(x,y)$

Consider $\frac{dy}{\tan y} = \frac{dz}{\tan z}$

$\Rightarrow \cot y\, dy = \cot z\, dz$

$\frac{\sin y}{\sin z} = C_2 = v(y,z)$

$\therefore$ Complete Solutions

$$u(x,y) = \frac{\sin x}{\sin y}$$

$$v(y,z) = \frac{\sin y}{\sin z}$$

2. $\frac{y^2z}{x}p + xzq = y^2$

Sol: Given $\frac{y^2z}{x}p + xzq = y^2$

$\Rightarrow y^2zp + x^2zq = y^2x$

It is in the form of Lagranger linear Equation: $Pp+Qq=R$

$\Rightarrow P = y^2z \quad Q = x^2z \quad R = y^2x$

Auxilary Equation: $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$\Rightarrow \frac{dx}{y^2z} = \frac{dy}{x^2z} = \frac{dz}{y^2x}$

By grouping method

Consider $\frac{dx}{y^2z} = \frac{dy}{x^2z}$

$\Rightarrow x^2dx = y^2dy$

Integrate on both sides

$\Rightarrow \int x^2dx = \int y^2dy$

$\Rightarrow \frac{x^3}{3} = \frac{y^3}{3} + c$

$\Rightarrow x^3 - y^3 = c_1 = u$

Consider $\frac{dx}{y^2z} = \frac{dz}{y^2x}$

$\Rightarrow xdx = zdz$

Integrate on both sides

$\Rightarrow \int xdx = \int zdz$

$\Rightarrow \frac{x^2}{2} = \frac{z^2}{2} + C$

$\Rightarrow x^2 - z^2 = c_2 = v$

$\therefore$ Solution $\phi(u, v) = 0$

$\Rightarrow \boxed{\phi(x^3 - y^3, x^2 - z^2) = 0}$

3. $y^2p - xyq = x(z-2y)$.

Auxiliary Equations: $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$\Rightarrow \frac{dx}{y^2} = \frac{dy}{-xy} = \frac{dz}{x(z-2y)}$

By grouping method

Consider

$\frac{dx}{y^2} = \frac{dy}{-xy}$

$\Rightarrow x\,dx = -y\,dy$

Integrate on both sides

$\Rightarrow \frac{x^2}{2} = \frac{-y^2}{2} + C$

$\Rightarrow x^2 + y^2 = c_1 = u$

Consider

$\frac{dy}{-xy} = \frac{dz}{x(z-2y)}$

$(z-2y)dy = -y\,dz$

Integrate on both sides

$\Rightarrow zy - y^2 = -zy + C$

$\Rightarrow 2zy - y^2 = c_2 = v$

$\therefore$ Solution $\phi(u,v) = \boxed{\phi(x^2+y^2, 2zy-y^2) = 0}$

4. $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$.

Auxilary Equation:

$$\frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)}$$

By Multipliers Method

Let $l_1 = \frac{1}{x^2}$, $m_1 = \frac{1}{y^2}$, $n_1 = \frac{1}{z^2}$

Now $l_1P + m_1Q + n_1R = \frac{1}{x^2}(x^2(y-z)) + \frac{y^2(z-x)}{y^2} + \frac{z^2(x-y)}{z^2}$

$= y - z + z - x + x - y$

$= 0 - (1)$

$$\Rightarrow \frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)} = \frac{l_1dx + m_1dy + n_1dz}{l_1P + m_1Q + n_1R}$$ From (1)

$$= \frac{l_1dx + m_1dy + n_1dz}{0}$$

$\Rightarrow l_1dx + m_1dy + n_1dz = 0$

$\Rightarrow \frac{1}{x^2}dx + \frac{1}{y^2}dy + \frac{1}{z^2}dz = 0$ Integrate on both sides

$\Rightarrow -\frac{1}{x} - \frac{1}{y} - \frac{1}{z} = c$

$\Rightarrow \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = c_1 = u$

Let $l_2 = \frac{1}{x}$, $m_2 = \frac{1}{y}$, $n_2 = \frac{1}{z}$

Now $l_2P + m_2Q + n_2R = \frac{x^2(y-z)}{x} + \frac{y^2(z-x)}{y} + \frac{z^2(x-y)}{z}$

$= xy - xz + yz - xy + xz - yz$

$= 0 - (2)$

$$\Rightarrow \frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)} = \frac{l_2dx + m_2dy + n_2dz}{l_2P + m_2Q + n_2R}$$ From ②

$$= \frac{l_2dx + m_2dy + n_2dz}{0}$$

$\Rightarrow l_2dx + m_2dy + n_2dz = 0$

$\Rightarrow \frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz = 0$ Integrating on both sides

$\Rightarrow \log x + \log y + \log z = c$

$\Rightarrow \log(xyz) = c$

$\Rightarrow xyz = c_2 = v$

$\therefore$ Solution $\phi(u, v) = \boxed{\phi\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}, xyz\right) = 0}$

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5. $p^2x^2 - q^2y^2 = z(x-y)$

Sol: Auxiliary Equations: $\frac{dx}{x^2} = \frac{dy}{-y^2} = \frac{dz}{z(x-y)}$

By grouping method

Consider: $\frac{dx}{x^2} = \frac{dy}{-y^2}$

Integrating on both sides

$\Rightarrow -\frac{1}{x} + c = \frac{1}{y}$

$\Rightarrow \frac{1}{x} + \frac{1}{y} = c = u$

By multipliers method

Let $l_1 = \frac{1}{x}$, $m = \frac{1}{y}$, $n = \frac{-1}{z}$

Now $lP + mQ + nR = \frac{1}{x} \times x^2 - \frac{1}{y} \times y^2 + \left(\frac{-1}{z}\right)(x-y)$

$= x - y - x + y$

$= 0 - (1)$

$\Rightarrow \frac{dx}{x^2} = \frac{dy}{-y^2} = \frac{dz}{z(x-y)} = \frac{l\,dx + m\,dy + n\,dz}{lP + mQ + nR}$ From (1)

$= \frac{l\,dx + m\,dy + n\,dz}{0}$

$\Rightarrow l\,dx + m\,dy + n\,dz = 0$

$\Rightarrow \frac{dx}{x} + \frac{dy}{y} - \frac{dz}{z} = 0$ Integrating on both sides

$\Rightarrow \log x + \log y - \log z = c$

$\Rightarrow \log \frac{xy}{z} = c$

$\Rightarrow \frac{xy}{z} = c_1 = V$

$\therefore$ Solution $\phi(u, v) = \boxed{\phi\left(\frac{1}{x} + \frac{1}{y}, \frac{xy}{z}\right) = 0}$

6. $(z^2-2yz-y^2)p+(xy+zx)q=(xy-zx)$

Sol: Auxiliary Equation: $\frac{dx}{z^2-2yz-y^2}=\frac{dy}{xy+zx}=\frac{dz}{xy-zx}$

By Grouping Method

Consider $\frac{dy}{xy+zx}=\frac{dz}{xy-zx}$

$\Rightarrow \frac{dy}{x(y+z)}=\frac{dz}{x(y-z)}$

$\Rightarrow (y-z)dy=(y+z)dz$

$\Rightarrow ydy-zdy=ydz+zdz$

$\Rightarrow ydy=ydz+zdy+zdz$

$\Rightarrow ydy=d(yz)+zdz$ Integrate on bs

$\Rightarrow y^2-2yz-z^2=c_1=u$

By Multipliers Method

let $l=x \quad m=y \quad n=z$

Now $lP+mQ+nR=0$ - (1)

$\Rightarrow \frac{dx}{z^2-2yz-y^2}=\frac{dy}{xy+zx}=\frac{dz}{xy-zx}=\frac{ldx+mdy+ndz}{lP+mQ+nR}$ From (1)

$\Rightarrow ldx+mdy+ndz=0$

$\Rightarrow xdx+ydy+zdz=0$ Integrate on bs

$\Rightarrow x^2+y^2+z^2=c_2=V$

$\therefore$ Solution $\phi(u,v)=$ $\boxed{\phi(y^2-2yz-z^2, x^2+y^2+z^2)=0}$

7. $(mz-ny)p+(nx-lz)q = ly-mx$

Sol: Auxiliary Equation

$$\frac{dx}{mz-ny} = \frac{dy}{nx-lz} = \frac{dz}{ly-mx}$$

By Multipliers method

Let $l_1 = x, m_1 = y, n_1 = z$

$\Rightarrow l_1P + m_1Q + n_1R = 0 \quad -(1)$

$$\Rightarrow \frac{dx}{mz-ny} = \frac{dy}{nx-lz} = \frac{dz}{ly-mx} = \frac{l_1dx+m_1dy+n_1dz}{l_1P+m_1Q+n_1R}$$

$$= \frac{l_1dx+m_1dy+n_1dz}{0} \quad \text{From (1)}$$

$\Rightarrow l_1dx+m_1dy+n_1dz = 0$

$\Rightarrow xdx+ydy+zdz = 0$ Integrate on both sides

$\Rightarrow x^2+y^2+z^2 = c_1 = u$

Let $l_2 = l, m_2 = m, n_2 = n$

$\Rightarrow l_2P + m_2Q + n_2R = 0 \quad -(2)$

$$\Rightarrow \frac{dx}{mz-ny} = \frac{dy}{nx-lz} = \frac{dz}{ly-mx} = \frac{l_2dx+m_2dy+n_2dz}{l_2P+m_2Q+n_2R}$$

$$= \frac{l_2dx+m_2dy+n_2dz}{0} \quad \text{From (2)}$$

$\Rightarrow l_2dx+m_2dy+n_2dz = 0$

$\Rightarrow ldx+mdy+ndz = 0$ Integrate on both sides

$\Rightarrow lx+my+nz = c = v$

$\therefore$ Solution $\phi(u,v) = \boxed{\phi(x^2+y^2+z^2, lx+my+nz) = 0}$

8. $(x^2-y^2-z^2)p + 2xyq = 2xz$

Sol Auxiliary Equations: $\frac{dx}{x^2-y^2-z^2} = \frac{dy}{2xy} = \frac{dz}{2xz}$

By grouping Method

Consider $\frac{dy}{2xy} = \frac{dz}{2xz}$ Integrate on both sides

$\Rightarrow \log y = \log z + C$

$\Rightarrow -\log y + \log z = C$

$\Rightarrow \log\left(\frac{z}{y}\right) = C$

$\Rightarrow \frac{z}{y} = c_1 = u$

By Multipliers Method

let $l=0, m=z, n=-y$

$\Rightarrow lp + mQ + nR = 0 \quad -(1)$

$\Rightarrow \frac{dx}{x^2-y^2-z^2} = \frac{dy}{2xy} = \frac{dz}{2xz} = \frac{l\,dx + m\,dy + n\,dz}{lp + mQ + nR}$ From (1)

$= \frac{l\,dx + m\,dy + n\,dz}{0}$

$\Rightarrow l\,dx + m\,dy + n\,dz = 0$

$\Rightarrow 0 + (z\,dy - y\,dz) \times \frac{z^2}{z^2} = 0$

$\Rightarrow \frac{z\,dy - y\,dz}{z^2} = \frac{0}{z^2}$

$\Rightarrow d\left(\frac{y}{z}\right) = 0$ Integrate on both sides

$\Rightarrow \frac{y}{z} = C = V$ $\therefore$ Solution $\phi(u, v)$

$= \boxed{\phi\left(\frac{z}{y}, \frac{y}{z}\right) = 0}$

9. $4r + 12s + 9t = 0$

Sol: Given, $4r + 12s + 9t = 0$

$\Rightarrow$ DE: $4\frac{\partial^2 z}{\partial x^2} + 12\frac{\partial^2 z}{\partial x \partial y} + 9\frac{\partial^2 z}{\partial y} = 0$

Operator Form: $(4D^2 + 12DD' + 9(D')^2)z = 0 = Q(x,y)$

$\Rightarrow F(D,D') = 0$

Auxiliary Equation: $F(m,1) = 0$

$\Rightarrow 4m^2 + 12m + 9 = 0$

$\Rightarrow m = -\frac{3}{2}, -\frac{3}{2}$ real and equal roots

$\therefore CF = \phi_1(y - \frac{3}{2}x) + x\phi_2(y - \frac{3}{2}x)$

$PI = 0 \quad (\because Q(x,y) = 0)$

$\therefore$ Solution $z = CF + PF = \phi_1(y - \frac{3}{2}x) + x\phi_2(y - \frac{3}{2}x)$

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10. $\frac{\partial^3 z}{\partial x^3} - 3\frac{\partial^3 z}{\partial x^2 \partial y} + 4\frac{\partial^3 z}{\partial y^3} = e^{x+2y}$

Sol: Operator form: $(D^3 - 3D^2D' + 4(D')^3)z = e^{x+2y} = Q(x,y)$

Auxiliary Equation: $F(m,1) = 0$

$\Rightarrow m^3 - 3m^2 + 4 = 0$

$\Rightarrow m = -1, 2, 2$ Roots are real

$\therefore CF = \phi_1(y - x) + \phi_2(y - 2x) + x\phi_3(y - 2x)$

$\phi(x,y) = e^{x+2y} = e^{ax+by} \Rightarrow a=1 \quad b=2$

$$PI = \frac{1}{f(D,D')} Q(x,y)$$

$$= \frac{1}{D^3 - 3D^2D' + 4(D')^3} e^{x+2y} \quad \text{put } D=1, D'=2$$

$$= \frac{e^{x+2y}}{27}$$

$\therefore$ General solution = CF + PI

$$z = \phi_1(y-x) + \phi_2(y-2x) + x\phi_3(y-2x) + \frac{e^{x+2y}}{27}$$

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11. $(D^2 - DD')z = \sin x \cos 2y$

Sol, $f(D,D')z = \sin x \cos 2y = Q(x,y)$

Auxiliary Equation: $f(m,1) = 0$

$\Rightarrow m^2 - m = 0$

$\Rightarrow m = 0, 1$

$\therefore CF = \phi_1(y) + \phi_2(y+x)$

$$Q(x,y) = \sin x \cos 2y \times \frac{2}{2} = \frac{\sin(x+2y) + \sin(x-2y)}{2}$$

$$PI = \frac{1}{f(D,D')} [Q(x,y)]$$

$$= \frac{1}{D^2 - DD'} \left[\frac{\sin(x+2y) + \sin(x-2y)}{2}\right]$$

$$= \frac{1}{2}\left[\frac{\sin(x+2y)}{D^2-DD'} + \frac{\sin(x-2y)}{D^2-DD'}\right]$$

let $D^2=-1, DD'=-2$ let $D^2=-1, DD'=2$

$$= \frac{1}{2}\left[\frac{\sin(x+2y)}{1} + \frac{\sin(x-2y)}{-3}\right]$$

$$PI = \frac{\sin(x+2y)}{2} - \frac{\sin(x-2y)}{6}$$

$\therefore$ Solution $=$ CF + PI

$$z = \phi_1(y) + \phi_2(y+x) + \frac{\sin(x+2y)}{2} - \frac{\sin(x-2y)}{6}$$

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12. $(D^3+D^2D'-DD'^2-(D')^3)z = 3\sin(x+y)$

Sol $F(D,D')z = Q(x,y)$

Auxilary Equation : $F(m,1)=0$

$\Rightarrow m^3+m^2-m-1=0$

$\Rightarrow m = 1, -1, -1$ real roots

$\therefore CF = \phi_1(y+x) + \phi_2(y-x) + x\phi_3(y-x)$

$Q(x,y) = 3\sin(x+y) = \sin(ax+by) \Rightarrow a=1, b=1$

$$PI = \frac{1}{F(D,D')}Q(x,y)$$

$$= \frac{1}{D^3+D^2D'-D(D')^2-(D')^3}\,3\sin(x+y)$$

let $D^2 = -a = -1$
$DD' = -ab = -1$
$(D')^2 = -b = -1$

$$= \frac{1}{-D-D'+D'+D'}\,3\sin(x+y) = \frac{3\sin(x+y)}{0}$$

Failure case ->.

Now $\frac{d}{dD}F(D,D') = 3D^2+2DD'-(D')^2-0$

$$PI = \frac{x}{3D^2+2DD'-(D')^2} 3\sin(x+y)$$

$$= \frac{x}{3(-1)+2(-1)-(-1)} 3\sin(x+y)$$

$$= -\frac{3}{4} x \sin(x+y)$$

$\therefore$ Solution = CF + PI

$$z = \phi_1(y+x) + \phi_2(y-x) + x\phi_3(y-x) - \frac{3}{4} x\sin(x+y)$$

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3. $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6\frac{\partial^2 z}{\partial y^2} = x+y$

Sol Operator Form: $(D^2 + DD' - 6(D')^2)z = x+y$

$$\Rightarrow f(D,D')z = Q(x,y)$$

Auxilary Equation: $f(m,1) = 0$

$$\Rightarrow m^2 + m - 6 = 0$$

$m = -3, 2$ Real roots

$\therefore$ CF = $\phi_1(y-3x) + \phi_2(y+2x)$

$Q(x,y) = x+y = ax+by \Rightarrow a=1, b=1$

Now $f(D,D') = f(1,1) = 1^2 + 1\times1 - 6\times1 = -4 \neq 0$

$\therefore$ $PI = \frac{1}{f(a,b)} \int\int Q(u)\,du\,du$ where $u = (x+y)$

$$\Rightarrow PI = \frac{-1}{4} \int\int u\,du\,du = -\frac{u^3}{24} = -\frac{(x+y)^3}{24}$$

$\therefore$ Solution = CF + PI

$$z = \phi_1(y-3x) + \phi_2(y+2x) - \frac{(x+y)^3}{24}$$

14. $\frac{\partial^3 z}{\partial x^3} - 2\frac{\partial^3 z}{\partial x^2 \partial y} = 2e^{2x} + 3x^2y$

Sol Operator Form: $(D^3 - 2D^2D')z = 2e^{2x} + 3x^2y$

$$f(D, D')z = Q(x, y)$$

Auxiliary Equation: $f(m, 1) = 0$

$\Rightarrow m^3 - 2m^2 = 0$

$\Rightarrow m = 0, 0, 2$ Real roots

$\therefore CF = \phi_1(y) + x\phi_2(y) + \phi_3(y + 2x)$

$$PI = \frac{1}{f(D, D')} Q(x, y)$$

$$= \frac{1}{D^3 - 2D^2D'}(2e^{2x}) + \frac{1}{D^3 - 2D^2D'} 3x^2y$$

$$= \frac{1}{2^3 - 2x2^2 \times 0} 2e^{2x} + \frac{3x^2y}{D^3\left[1 - 2\frac{D'}{D}\right]}$$

$$= \frac{e^{2x}}{4} + \frac{3x^2y}{D^3}\left[1 - 2\frac{D'}{D}\right]^{-1}$$

$$= \frac{e^{2x}}{4} + \frac{3x^2y}{D^3}\left[1 + \frac{2D'}{D} + \frac{4(D')^2}{D^2} + \cdots\right]$$

$$= \frac{e^{2x}}{4} + \frac{3}{D^3}\left[x^2y + \frac{3x^2}{D} + 0\right] \quad \left(\frac{1}{D} = \int dx\right)$$

$$= \frac{e^{2x}}{4} + \frac{3}{D^3}\left[x^2y + x^3\right]$$

$$= \frac{e^{2x}}{4} + \frac{3x^5y}{60} + \frac{3x^6}{120} = \frac{e^{2x}}{4} + \frac{x^5y}{20} + \frac{x^6}{40}$$

15. $(D^2 - DD' - 2(D')^2)z = (y-1)e^x$

Sol Auxilary Equation: $f(m,1) = 0$

$\Rightarrow m^2 - m - 2 = 0$

$m = 2, -1$

$CF = \phi_1(y+2x) + \phi_2(y-x)$

$PI = \frac{1}{f(D,D')} Q(x,y)$

$= \frac{1}{D^2 - DD' - 2(D')^2} e^x (y-1)$

$= e^x \frac{1}{(D+1)^2 - (D+1)D' - 2(D')^2} (y-1)$

$= e^x \frac{1}{D^2 + 1 + 2D - DD' - D' - 2(D')^2} (y-1)$

$\therefore = e^x \left[\frac{y}{D^2 + 1 + 2D - DD' - D' - 2(D')^2} - \frac{1}{D^2 + 1 + 2D - DD' - 2(D')^2} \right]$

using $x^0 y^1$ Formula

using e^{ax+by} Formula, $a = 0$, $b = 0$

$= e^x \left[y\left[1 + (D^2 + 2D - DD' - D' - 2(D')^2)\right]^{-1} - \frac{1}{1} \right]$

only upto 1 term Dash to y^1

$= e^x (y[1 + 0 + 0 - - -] - 1)$

$= e^x (y-1)$

$e^{ax+by} V(x,y)$

$PI = e^{ax+by} \frac{1}{f(D+a, D'+b)} V(x,y)$

$\therefore$ Solution $\boxed{z = \phi_1(y+2x) + \phi_2(y-x) + e^x(y-1)}$

16. $(D^2+2DD'-8(D')^2)z = \sqrt{2x+3y}$

Sol Auxilary Form: $f(m,1)=0$

$\Rightarrow m^2+2m-8=0$

$m=-4,2$ Real Roots

$CF = \phi_1(y-4x)+\phi_2(y+2x)$

$Q(x,y) = \sqrt{2x+3y} = (ax+by)^{1/2} \Rightarrow a=2, b=3$

$f(2,3) = 2^2+2\times2\times3-8\times9 = -56 \neq 0$

$\therefore PI = \frac{1}{f(2,3)} \iint u^{1/2}\,du\,du$ where $u = 2x+3y$

$u^{1/2} = Q(x,y)$

$= \frac{-1}{56} \times \frac{4}{15}(2x+3y)^{5/2}$

$= \frac{-(2x+3y)^{5/2}}{15}$

$\therefore$ Solution $z = \phi_1(y-4x)+\phi_2(y+2x)-\frac{(2x+3y)^{5/2}}{15}$

UNIT-4

1. Find Angle between Surfaces $x^2+y^2+z^2=9$ & $z=x^2+y^2-3$ at $(2,-1,2)$.

Sol. Let Surfaces, $\phi_1 = x^2+y^2+z^2-9=0$ Point $P(2,-1,2)$

$\phi_2 = x^2+y^2-z-3=0$

Normal unit vector to the surface $\phi_1 = \nabla\phi_1$

$$\Rightarrow \nabla\phi_1 = \frac{\partial\phi_1}{\partial x}\bar{i} + \frac{\partial\phi_1}{\partial y}\bar{j} + \frac{\partial\phi_1}{\partial z}\bar{k}$$

$$= 2x\bar{i} + 2y\bar{j} + 2z\bar{k}$$

Normal vector at point P

$$n_1 = (\nabla\phi_1)_P = 2(2)\bar{i} + 2(-1)\bar{j} + 2(2)\bar{k}$$

$$n_1 = 4\bar{i} - 2\bar{j} + 4\bar{k} \quad \& \quad |\bar{n}_1| = \sqrt{4^2+2^2+4^2} = 6$$

Normal unit vector to the surface $\phi_2 = \nabla\phi_2$

$$\Rightarrow \nabla\phi_2 = \frac{\partial\phi_2}{\partial x}\bar{i} + \frac{\partial\phi_2}{\partial y}\bar{j} + \frac{\partial\phi_2}{\partial z}\bar{k}$$

$$= 2x\bar{i} + 2y\bar{j} - \bar{k}$$

Normal vector at point P

$$n_2 = (\nabla\phi_2)_P = 2(2)\bar{i} + 2(-1)\bar{j} - (1)\bar{k}$$

$$n_2 = 4\bar{i} - 2\bar{j} - \bar{k} \quad \& \quad |\bar{n}_2| = \sqrt{4^2+2^2+1^2} = \sqrt{21}$$

Let θ = Angle between ϕ_1 & ϕ_2.

WKT $\cos\theta = \dfrac{\bar{n}_1 \cdot \bar{n}_2}{|\bar{n}_1||\bar{n}_2|}$

$$= \frac{(4\bar{i}-2\bar{j}+4\bar{k})\cdot(4\bar{i}-2\bar{j}-\bar{k})}{6*\sqrt{21}} = \frac{16}{6\sqrt{21}} = \frac{8}{3\sqrt{21}}$$

$$\boxed{\theta = \cos^{-1}\left(\frac{8}{3\sqrt{21}}\right)}$$

2. Find Angle between the surfaces $xy = z^2$ at $(4,1,2)$ & $(3,3,-3)$

Sol. Let surface $\phi_1 = xy - z^2 = 0$

Points $P = (4,1,2)$ & $Q(3,3,-3)$

Normal unit vector of $\phi_1 = \nabla\phi_1$

$$\Rightarrow \nabla\phi_1 = \frac{\partial\phi_1}{\partial x}\bar{i} + \frac{\partial\phi_1}{\partial y}\bar{j} + \frac{\partial\phi_1}{\partial z}\bar{k}$$

$$= y\bar{i} + x\bar{j} - 2z\bar{k}$$

Normal unit vector at P

$$\bar{n}_1 = (\nabla\phi_1)_P = \bar{i} + 4\bar{j} - 2\times 2\bar{k} = \bar{i} + 4\bar{j} - 4\bar{k}$$

$$|\bar{n}_1| = \sqrt{1+4^2+4^2} = \sqrt{33}$$

Normal unit vector at Q

$$\bar{n}_2 = (\nabla\phi_1)_Q = 3\bar{i} + 3\bar{j} - 2\times 3\bar{k} = 3\bar{i} + 3\bar{j} + 6\bar{k}$$

$$|\bar{n}_1| = \sqrt{3^2+3^2+6^2} = \sqrt{54}$$

Let θ = Angle between the surfaces

$$\text{WKT, } \cos\theta = \frac{\bar{n}_1 \cdot \bar{n}_2}{|\bar{n}_1||\bar{n}_2|} = \frac{(\bar{i}+4\bar{j}-4\bar{k})\cdot(3\bar{i}+3\bar{j}+6\bar{k})}{\sqrt{33}\times\sqrt{54}}$$

$$= \frac{-9\sqrt{9}\sqrt{3}\sqrt{1}}{\sqrt{33}\sqrt{54}\sqrt{6}} = \frac{-1}{\sqrt{22}}$$

$$\therefore \theta = \cos^{-1}\left(\frac{-1}{\sqrt{22}}\right)$$

3. Find the values of a & b such that surfaces $ax^2 - byz = (a+2)x$ and $4x^2y + z^3 = 4$ cut orthogonally at $(1,-1,2)$.

Sol. Let surfaces, $\phi_1 = ax^2 - byz - (a+2)x = 0$ Point $P = (1,-1,2)$

$$\phi_2 = 4x^2y + z^3 - 4 = 0$$

Normal unit vector of $\phi_1 = \nabla\phi_1$

$$\Rightarrow \nabla\phi_1 = \frac{\partial\phi_1}{\partial x}\bar{i} + \frac{\partial\phi_1}{\partial y}\bar{j} + \frac{\partial\phi_1}{\partial z}\bar{k} = (2ax - a - 2)\bar{i} - bz\bar{j} - by\bar{k}$$

Normal unit vector at P

$$\bar{n}_1 = (\nabla\phi_1)_P = (a-2)\bar{i} - 2b\bar{j} + b\bar{k}$$

Normal unit vector of $\phi_2 = \nabla\phi_2$

$$\Rightarrow \nabla\phi_2 = \frac{\partial\phi_2}{\partial x}\bar{i} + \frac{\partial\phi_2}{\partial y}\bar{j} + \frac{\partial\phi_2}{\partial z}\bar{k} = 8xy\,\bar{i} + 4x^2\bar{j} + 3z^2\bar{k}$$

Normal unit vector at P

$$\bar{n}_2 = (\nabla\phi_2)_P = -8\bar{i} + 4\bar{j} + 12\bar{k}$$

Given, ϕ_1 & ϕ_2 cut orthogonally at P.

then, $\bar{n}_1 \cdot \bar{n}_2 = 0$

$$\Rightarrow (a-2)\bar{i} - 2b\bar{j} + b\bar{k}) \cdot (-8\bar{i} + 4\bar{j} + 12\bar{k}) = 0$$

$$\Rightarrow -8a + 16 - 8b + 12b = 0$$

$$\Rightarrow -8a + 4b + 16 = 0$$

$$\Rightarrow 2a - b - 4 = 0 \quad -(1)$$

Since ϕ_1 passes through P, Substitute P in ϕ_1

$$\Rightarrow a(1) - b(-1)(2) - (a+2)(1) = 0$$

$$\Rightarrow a + 2b - a - 2 = 0$$

$$\Rightarrow \boxed{b = 1}$$ Substitute in (1)

$$\Rightarrow \boxed{a = \frac{5}{2}}$$

4. Find Directional Derivative of $f(x,y,z) = xy^3 + yz^3$ at $(2,-1,1)$ in direction of the vector $\bar{i}+2\bar{j}+2\bar{k}$.

Sol. Given,

$$f(x,y,z) = xy^3 + yz^3$$

$$\nabla f = \frac{\partial f}{\partial x}\bar{i} + \frac{\partial f}{\partial y}\bar{j} + \frac{\partial f}{\partial z}\bar{k} = y^3\bar{i} + (3xy^2+z^3)\bar{j} + 3yz^2\bar{k}$$

At point $P = (2,-1,1)$

$$(\nabla f)_P = (-1)^3\bar{i} + (3\times 2\times 1 + 1^3)\bar{j} + (3\times -1\times 1)\bar{k}$$

$$= -\bar{i} + 7\bar{j} - 3\bar{k}$$

Unit vector in the direction of $\bar{i}+2\bar{j}+2\bar{k}$

$$\bar{n} = \frac{\bar{i}+2\bar{j}+2\bar{k}}{|\sqrt{1+2^2+2^2}|} = \frac{\bar{i}+2\bar{j}+2\bar{k}}{3}$$

Directional Derivative of f at P in direction of $\bar{i}+2\bar{j}+3\bar{k}$

$$\Rightarrow D_n = \bar{n}.(\nabla f)_P$$

$$= \frac{1}{3}(\bar{i}+2\bar{j}+2\bar{k}).(-\bar{i}+7\bar{j}-3\bar{k})$$

$$= \frac{-1+14-6}{3}$$

$$\boxed{D_n = \frac{7}{3}}$$

5. Find Directional derivative of $\phi(x,y,z) = xy^2 + yz^3$ at $(2,-1,1)$ in the direction of normal to surface $x\log z - y^2 = -4$ at $(-1,2,1)$

Sol. Given, surface $\phi = xy^2 + yz^3$

$$\nabla\phi = \frac{\partial\phi}{\partial x}\bar{i} + \frac{\partial\phi}{\partial y}\bar{j} + \frac{\partial\phi}{\partial z}\bar{k} = y^2\bar{i} + (2xy + z^3)\bar{j} + 3yz^2\bar{k}$$

At point $(2,-1,1)$

$$(\nabla\phi)_p = 1\times\bar{i} + (2\times2\times-1*1)\bar{j} + 3(-1)\times1\,\bar{k}$$

$$= \bar{i} - 3\bar{j} - 3\bar{k}$$

Surface $f = x\log z - y^2 = -4$

Normal unit vector to f at point $(-1,2,1)$

$$(\nabla f)_p = \left(\frac{\partial f}{\partial x}\bar{i} + \frac{\partial f}{\partial y}\bar{j} + \frac{\partial f}{\partial z}\bar{k}\right)_p$$

$$= \left(\log z\,\bar{i} - 2y\bar{j} + \frac{x}{z}\bar{k}\right)_{(-1,2,1)}$$

$$= 0\bar{i} - 4\bar{j} - \bar{k}$$

Unit vector in the direction of $(\nabla f)_p$

$$\bar{n} = \frac{(\nabla f)_p}{|(\nabla f)_p|} = \frac{-4\bar{j} - \bar{k}}{\sqrt{4^2+1}} = \frac{-4\bar{j} - \bar{k}}{\sqrt{17}}$$

Directional Derivative of ϕ in the direction of normal to f

$$D_n = \bar{n}.(\nabla\phi)_p$$

$$= \frac{1}{\sqrt{17}}(-4\bar{j} - \bar{k}).(\bar{i} - 3\bar{j} - 3\bar{k}) = \frac{12+3}{\sqrt{17}}$$

$$\boxed{D_n = \frac{15}{\sqrt{17}}}$$

6. Find D_n of $x^2 - y^2 + 2z^2$ at $P(1,2,3)$ in the direction of PQ where $Q(5,0,4)$. Also find the magnitude of maximum directional derivative.

Sol. Let $f \equiv x^2 - y^2 + 2z^2$

$$\nabla f = \frac{\partial f}{\partial x}\bar{i} + \frac{\partial f}{\partial y}\bar{j} + \frac{\partial f}{\partial z}\bar{k} = 2x\bar{i} - 2y\bar{j} + 4z\bar{k}$$

At point $P(1,2,3)$

$$(\nabla f)_P = 2\bar{i} - 4\bar{j} + 12\bar{k}$$

Given, points $P(1,2,3)$ & $Q(5,0,4)$

positional vectors of P & Q with respect to origin O

$\Rightarrow \bar{OP} = \bar{i} + 2\bar{j} + 3\bar{k}$ $\quad \bar{OQ} = 5\bar{i} + 4\bar{j}$

WKT $\bar{PQ} = \bar{OQ} - \bar{OP}$

$$\bar{PQ} = 4\bar{i} - 2\bar{j} + \bar{k}$$

Unit vector in the direction of $\bar{PQ}$

$$\bar{n} = \frac{\bar{PQ}}{|\bar{PQ}|} = \frac{4\bar{i} - 2\bar{j} + \bar{k}}{\sqrt{4^2 + 2^2 + 1^2}} = \frac{4\bar{i} - 2\bar{j} + \bar{k}}{\sqrt{21}}$$

Directional ~~vector~~ Derivative of f at P in direction of PQ

$$D_n = \bar{n} \cdot (\nabla f)_P$$

$$= \frac{1}{\sqrt{21}}(4\bar{i} - 2\bar{j} + \bar{k}) \cdot (2\bar{i} - 4\bar{j} + 12\bar{k}) = \frac{28}{\sqrt{21}} = 2\sqrt{\frac{7}{3}}$$

$$\boxed{D_n = 2\sqrt{\frac{7}{3}}}$$

Magnitude of Maximum $D_n = |(\nabla f)_P| = \sqrt{2^2 + 4^2 + 12^2} = \sqrt{164}$

$$= \boxed{2\sqrt{41}}$$

7. Find the directional derivative of $xy^2+yz^2+zx^2$ along the tangent to the curve $x=t, y=t^2, z=t^3$ at $(1,1,1)$

Sol: Let $\phi = xy^2+yz^2+zx^2$

Curve $x=t, y=t^2, z=t^3$ Point $P(x,y,z)=(1,1,1)$

$$\nabla\phi = \frac{\partial\phi}{\partial x}\bar{i} + \frac{\partial\phi}{\partial y}\bar{j} + \frac{\partial\phi}{\partial z}\bar{k} = (y^2+2xz)\bar{i} + (2xy+z^2)\bar{j} + (2yz+x^2)\bar{k}$$

At $P(1,1,1)$

$$(\nabla\phi)_P = 3\bar{i}+3\bar{j}+3\bar{k}$$

Let $\bar{r}$ be the positional vector of of any point on curve

$$\bar{r} = x\bar{i}+y\bar{j}+z\bar{k} \qquad x=t, y=t^2, z=t^3$$

$$\bar{r} = t\bar{i}+t^2\bar{j}+t^3\bar{k}$$

WKT Tangent vector to the curve $= \frac{d\bar{r}}{dt} = \bar{T}$

$$\Rightarrow \frac{d\bar{r}}{dt} = \frac{d(t)}{dt}\bar{i} + \frac{d(t^2)}{dt}\bar{j} + \frac{d(t^3)}{dt}\bar{k}$$

$$\Rightarrow \bar{T} = \bar{i}+2t\bar{j}+3t^2\bar{k}$$

At $P(x,y,z)=(1,1,1)$ From curve

$P(t,t^2,t^3)=(1,1,1)$

$\therefore t=1$

$$\Rightarrow \bar{T} = \bar{i}+2\bar{j}+3\bar{k}$$

Unit vector in the direction of $\bar{T} = \bar{n} = \frac{\bar{T}}{|\bar{T}|} = \frac{\bar{i}+2\bar{j}+3\bar{k}}{\sqrt{14}}$

Directional Derivative of ϕ along $\bar{T}$

$$D_n = \bar{n}\cdot(\nabla\phi)_P = \frac{1}{\sqrt{14}}(\bar{i}+2\bar{j}+3\bar{k})\cdot(3\bar{i}+3\bar{j}+3\bar{k}) = \frac{3+6+9}{\sqrt{14}} = \frac{18}{\sqrt{14}} = \frac{9\sqrt{2}}{\sqrt{7}}$$

$$\boxed{D_n = 9\sqrt{\frac{2}{7}}}$$

8. Temperature of points in a space is given by $T = x^2+y^2-z$, a mosquito is located at $(1,1,2)$ desires to fly to a hotter place. In what direction it should move. Also Find the max rate of change in the temperature

Sol: Given,

Temperature function $T = x^2+y^2-z$

mosquito is at $P(1,1,2)$

Rate of change of temperature = Temperature gradient

$$\Rightarrow \nabla T = \frac{\partial T}{\partial x}\bar{i} + \frac{\partial T}{\partial y}\bar{j} + \frac{\partial T}{\partial z}\bar{k} = 2x\bar{i} + 2y\bar{j} - \bar{k}$$

At point $P(1,1,2)$

$$\Rightarrow (\nabla T)_P = 2\bar{i} + 2\bar{j} - \bar{k} = (2,2,-1)$$

∴ The mosquito should fly from P in the direction of the directional vector $2\bar{i}+2\bar{j}-\bar{k}$

Maximum Rate of change in temperature

$$= |(\nabla T)_P|$$

$$= |\sqrt{2^2+2^2+1^2}|$$

$$= \boxed{3}$$

9. Find the div $\bar{F}$ & curl $\bar{F}$, $\bar{F}$ = grad $(x^3+y^3+z^3-3xyz)$

Sol Let $\phi = x^3+y^3+z^3-3xyz$

Given, $\bar{F} = \nabla\phi = \nabla(x^3+y^3+z^3-3xyz)$

$$= \frac{\partial\phi}{\partial x}\bar{i} + \frac{\partial\phi}{\partial y}\bar{j} + \frac{\partial\phi}{\partial z}\bar{k}$$

$$\bar{F} = (3x^2-3yz)\bar{i} + (3y^2-3xz)\bar{j} + (3z^2-3xy)\bar{k}$$

WKT if $\bar{F} = f_1\bar{i} + f_2\bar{j} + f_3\bar{k}$ then

$$\text{div}\,\bar{F} = \nabla\cdot\bar{F} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

Here $f_1 = 3x^2-3yz$, $f_2 = 3y^2-3xz$, $f_3 = 3z^2-3xy$

$$\Rightarrow \boxed{\text{div}\,\bar{F} = 6x+6y+6z}$$

WKT $\text{curl}\,\bar{F} = \nabla\times\bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$

$$\Rightarrow \text{curl}\,\bar{F} = \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}\right)\bar{i} - \left(\frac{\partial f_3}{\partial x} - \frac{\partial f_1}{\partial z}\right)\bar{j} + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y}\right)\bar{k}$$

$$= ((0-0)-(0-0))\bar{i} - ((0-0)-(0-0))\bar{j} + ((0-0)-(0-0))\bar{k}$$

$$= 0\bar{i} + 0\bar{j} + 0\bar{k}$$

$$\boxed{\text{curl}\,\bar{F} = \bar{0}}$$

10. Find div $\bar{F}$ & curl $\bar{F}$ at $(1,1,-1)$, $\bar{F} = xy^2\bar{i} + 2x^2yz\bar{j} - 3yz^2\bar{k}$

Sol: Given, $\bar{F} = xy^2\bar{i} + 2x^2yz\bar{j} - 3yz^2\bar{k}$

Compare with $\bar{F} = F_1\bar{i} + F_2\bar{j} + F_3\bar{k}$

$F_1 = xy^2 \quad F_2 = 2x^2yz \quad F_3 = -3yz^2$

$$\text{div}\,\bar{F} = \nabla\cdot\bar{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

$$\text{div}\,\bar{F} = y^2 + 2x^2z - 6yz$$

At point $(1,1,-1)$

$$\text{div}\,\bar{F} = (1)^2 + 2(1)^2(-1) - 6(1)(-1)$$

$$\boxed{(\text{div}\,\bar{F})_{(1,1,-1)} = 9}$$

$$\text{Curl}\,\bar{F} = \nabla\times\bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

$$= \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right)\bar{i} - \left(\frac{\partial F_3}{\partial x} - \frac{\partial F_1}{\partial z}\right)\bar{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)\bar{k}$$

$$= (-3z^2 - 2x^2y)\bar{i} - (0-0)\bar{j} + (4xyz - 2xy)\bar{k}$$

At $(1,1,-1)$

$$= (-3\times1 - 2\times1\times1)\bar{i} + (4\times1\times1\times-1 - 2\times1\times1)\bar{k}$$

$$\boxed{(\text{Curl}\,\bar{F})_{(1,1,-1)} = -5\bar{i} - 6\bar{k}}$$

11. Prove that $\operatorname{div}(r^n \bar{r}) = (n+3) r^n$. Show that $\frac{\bar{r}}{r^3}$ is solenoidal

Sol. Let $\bar{F} = r^n \bar{r} = r^n x \bar{i} + r^n y \bar{j} + r^n z \bar{k}$

$\Rightarrow F_1 = r^n x, \; F_2 = r^n y, \; F_3 = r^n z$

$$\operatorname{div} \bar{F} = \nabla \cdot \bar{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

$$= \frac{\partial (r^n x)}{\partial x} + \frac{\partial (r^n y)}{\partial y} + \frac{\partial (r^n z)}{\partial z}$$

Consider: $\frac{\partial}{\partial x}(r^n x) = n r^{n-1} \frac{\partial r}{\partial x} x + r^n (1)$

$$= \left(n r^{n-1} \frac{x}{r} \right) x + r^n$$

$$= n r^{n-2} x^2 + r^n$$

Similarly $\frac{\partial}{\partial y}(r^n y) = n r^{n-2} y^2 + r^n$

$$\frac{\partial}{\partial z}(r^n z) = n r^{n-2} z^2 + r^n$$

$\Rightarrow \operatorname{div} \bar{F} = n r^{n-2} x^2 + n r^{n-2} y^2 + n r^{n-2} z^2 + r^n + r^n + r^n$

$= n r^{n-2} (x^2 + y^2 + z^2) + 3 r^n \quad (\because \bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$, $|\bar{r}|^2 = x^2 + y^2 + z^2)$

$= n r^{n-2} r^2 + 3 r^n$

$= n r^n + 3 r^n$

$= (n+3) r^n$

$\therefore \operatorname{div}(r^n \bar{r}) = (n+3) r^n$

Let $\bar{F} = \frac{\bar{r}}{r^3} = \frac{x\bar{i} + y\bar{j} + z\bar{k}}{r^3}$

$\Rightarrow F_1 = \frac{x}{r^3}, F_2 = \frac{y}{r^3}, F_3 = \frac{z}{r^3}$

$\text{div}\,\bar{F} = \nabla \cdot \bar{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$

$= \frac{\partial}{\partial x}\left(\frac{x}{r^3}\right) + \frac{\partial}{\partial y}\left(\frac{y}{r^3}\right) + \frac{\partial}{\partial z}\left(\frac{z}{r^3}\right)$

Consider $\frac{\partial}{\partial x}\left(\frac{x}{r^3}\right) = \frac{r^3(1) - x \times 3r^2\left(\frac{\partial r}{\partial x}\right)}{r^6} = r^{-3} - \frac{3x \times \frac{x}{r} \times x}{r^5}$

$\frac{\partial}{\partial x}\left(\frac{x}{r^3}\right) = r^{-3} - 3x^2 r^{-5}$

Similarly $\frac{\partial}{\partial y}\left(\frac{y}{r^3}\right) = r^{-3} - 3y^2 r^{-5}$

$\frac{\partial}{\partial z}\left(\frac{z}{r^3}\right) = r^{-3} - 3z^2 r^{-5}$

$\Rightarrow \text{div}\,\bar{F} = r^{-3} + r^{-3} + r^{-3} - 3x^2 r^{-5} - 3y^2 r^{-5} - 3z^2 r^{-5}$

$= 3r^{-3} - 3r^{-5}(x^2 + y^2 + z^2)$

$= 3r^{-3} - 3r^{-3} x r$

$= 3r^{-3} - 3r^{-3} = 0$

$$\boxed{\text{div}\left(\frac{\bar{r}}{r^3}\right) = 0}$$

WKT if $\text{div}\,F = 0$ then F is called solenoidal

$\therefore \frac{\bar{r}}{r^3}$ is Solenoidal

12. Show that $\bar{F} = 2xyz^2\bar{i} + (x^2z^2 + z\cos yz)\bar{j} + (2x^2yz + y\cos yz)\bar{k}$ is irrotational and find scalar potential function.

Sol: Given, $\bar{F} = 2xyz^2\bar{i} + (x^2z^2 + z\cos yz)\bar{j} + (2x^2yz + y\cos yz)\bar{k}$

$f_1 = 2xyz^2 \quad f_2 = x^2z^2 + z\cos yz \quad f_3 = 2x^2yz + y\cos yz$

$$\text{curl } \bar{F} = \nabla \times \bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$$

$$= \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}\right)\bar{i} - \left(\frac{\partial f_3}{\partial x} - \frac{\partial f_1}{\partial z}\right)\bar{j} + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y}\right)\bar{k}$$

$$\frac{\partial f_3}{\partial y} = 2x^2z + zy(-\sin yz) + \cos yz = 2x^2z - zy\sin yz + \cos yz$$

$$\frac{\partial f_2}{\partial z} = 2x^2z + yz(-\sin yz) + \cos yz = 2x^2z - yz\sin yz + \cos yz$$

$$\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} = 0 \quad -①$$

$$\frac{\partial f_3}{\partial x} = 4xyz + 0 \qquad \frac{\partial f_1}{\partial z} = 4xyz + 0 \qquad \frac{\partial f_3}{\partial x} - \frac{\partial f_1}{\partial z} = 0 \quad -②$$

$$\frac{\partial f_2}{\partial x} = 2xz^2 + 0 \qquad \frac{\partial f_1}{\partial y} = 2xz^2 + 0 \qquad \frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} = 0 \quad -③$$

$\Rightarrow \text{curl } \bar{F} = 0\bar{i} + 0\bar{j} + 0\bar{k}$ From ①, ②, ③

$\text{curl } \bar{F} = \bar{0}$

WKT If $\text{curl } \bar{F} = \bar{0}$ then $\bar{F}$ is irrotational

$\therefore \bar{F}$ is irrotational $\Leftrightarrow \exists$ a scalar potential function ϕ such that $\bar{F} = \nabla\phi$

$$\Rightarrow f_1\bar{i} + f_2\bar{j} + f_3\bar{k} = \frac{\partial \phi_1}{\partial x}\bar{i} + \frac{\partial \phi_2}{\partial y}\bar{j} + \frac{\partial \phi_3}{\partial z}\bar{k}$$

$$\Rightarrow \frac{\partial \phi_1}{\partial x} = f_1 \qquad \frac{\partial \phi_2}{\partial y} = f_2 \qquad \frac{\partial \phi_3}{\partial z} = f_3$$

$\Rightarrow \frac{\partial \phi_1}{\partial x} = 2xyz^2$

Integrating on both sides

$\Rightarrow \phi_1 = \int 2xyz^2\,dx$

$\phi_1 = x^2yz^2$

$\Rightarrow \frac{\partial \phi_2}{\partial y} = x^2z^2 + z\cos yz$

Integrating on both sides

$\Rightarrow \phi_2 = \int (x^2z^2 + z\cos yz)\,dy$

$= x^2yz^2 + \frac{z\sin yz}{z}$

$\phi_2 = x^2yz^2 + \sin yz$

$\Rightarrow \frac{\partial \phi_3}{\partial z} = 2x^2yz + y\cos yz$

Integrating

$\Rightarrow \phi_3 = \int (2x^2yz + y\cos yz)\,dz$

$\phi_3 = x^2yz^2 + \sin yz$

$\Rightarrow \phi = \phi_1 + \phi_2 + \phi_3$ (Write common terms once)

$\boxed{\phi = x^2yz^2 + \sin yz + c}$ is the scalar ~~point~~ potential function of $\bar{f}$

13. Show that Vector Field $\bar{F} = (x^2 - yz)\bar{i} + (y^2 - xz)\bar{j} + (z^2 - xy)\bar{k}$ is irrotational and hence find the scalar Potential Function

Sol Given, $\bar{F} = (x^2 - yz)\bar{i} + (y^2 - xz)\bar{j} + (z^2 - xy)\bar{k}$

$\Rightarrow f_1 = x^2 - yz$, $f_2 = y^2 - xz$, $f_3 = z^2 - xy$

If f is irrotational then curl $\bar{F} = \bar{0}$

$$\text{Curl } \bar{F} = \nabla \times \bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$$

$$= \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}\right)\bar{i} - \left(\frac{\partial f_3}{\partial x} - \frac{\partial f_1}{\partial z}\right)\bar{j} + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y}\right)\bar{k}$$

$\frac{\partial f_3}{\partial y} = 0 - x = -x$ | $\frac{\partial f_2}{\partial z} = 0 - x = -x$ | $\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} = 0$ — (1)

$\frac{\partial f_3}{\partial x} = 0 - y = -y$ | $\frac{\partial f_1}{\partial z} = 0 - y = -y$ | $\frac{\partial f_3}{\partial x} - \frac{\partial f_1}{\partial z} = 0$ — (2)

$\frac{\partial f_2}{\partial x} = 0 - z = -z$ | $\frac{\partial f_1}{\partial y} = 0 - z = -z$ | $\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} = 0$ — (3)

$\Rightarrow \text{Curl } \bar{F} = 0\bar{i} + 0\bar{j} + 0\bar{k}$

$\text{Curl } \bar{F} = \bar{0}$

$\therefore \bar{F}$ is irrotational and $\exists$ a scalar potential function ϕ such that $\bar{F} = \nabla\phi$

$$\Rightarrow f_1\bar{i} + f_2\bar{j} + f_3\bar{k} = \frac{\partial \phi_1}{\partial x}\bar{i} + \frac{\partial \phi_2}{\partial y}\bar{j} + \frac{\partial \phi_3}{\partial z}\bar{k}$$

$\Rightarrow f_1 = \frac{\partial \phi_1}{\partial x}$

$\Rightarrow \frac{\partial \phi_1}{\partial x} = x^2 - yz$

Integrating on both sides

$\Rightarrow \phi_1 = \int (x^2 - yz)\,dx$

$\Rightarrow \phi_1 = \frac{x^3}{3} - xyz$

Similarly

$\phi_2 = \frac{y^3}{3} - xyz$

$\phi_3 = \frac{z^3}{3} - xyz$

$\Rightarrow \phi = \phi_1 + \phi_2 + \phi_3 + C$ (write common terms only once)

$\Rightarrow \boxed{\phi = \frac{x^3}{3} + \frac{y^3}{3} + \frac{z^3}{3} - xyz + C}$

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14. Find a, b, c so that $\bar{A} = (x+2y+az)\bar{i} + (bx-3y-z)\bar{j} + (4x+cy+2z)\bar{k}$ is irrotational and find scalar potential function

Sol Given, $\bar{A} = (x+2y+az)\bar{i} + (bx-3y-z)\bar{j} + (4x+cy+2z)\bar{k}$

$\bar{A}$ is irrotational

$\Rightarrow \text{curl } \bar{A} = \bar{0}$

$\Rightarrow \bar{0} = \nabla \times \bar{A} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$

In $\bar{A}$

$f_1 = x + 2y + az$

$f_2 = bx - 3y - z$

$f_3 = 4x + cy + 2z$

$\Rightarrow \bar{0} = \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}\right)\bar{i} - \left(\frac{\partial f_3}{\partial x} - \frac{\partial f_1}{\partial z}\right)\bar{j} + \left(\frac{\partial f_2}{\partial y} - \frac{\partial f_1}{\partial x}\right)\bar{k}$

$= [0 + c + 0 - (0 - 0 - 1)]\bar{i} - [4 + 0 + 0 - (0 + 0 + a)]\bar{j} + [b - 0 - 0 - (0 + 2 + 1)]\bar{k}$

$\Rightarrow 0\bar{i} + 0\bar{j} + 0\bar{k} = (c+1)\bar{i} + (a-4)\bar{j} + (b-2)\bar{k}$

Comparing both sides $c + 1 = 0 \quad a - 4 = 0 \quad b - 2 = 0$

$\Rightarrow \boxed{a=4} \quad \boxed{b=2} \quad \boxed{c=-1}$

$\because A$ is irrotational $\exists$ a scalar potential function ϕ where

$\bar{A} = \nabla\phi$

$\Rightarrow f_1\bar{i} + f_2\bar{j} + f_3\bar{k} = \frac{\partial\phi_1}{\partial x} + \frac{\partial\phi_2}{\partial y} + \frac{\partial\phi_3}{\partial z}$

$\Rightarrow \frac{\partial\phi_1}{\partial x} = f_1$

$\Rightarrow \frac{\partial\phi_1}{\partial x} = x+2y+4z$

Integrating on bs

$\Rightarrow \phi_1 = \int (x+2y+4z)dx$

$\phi_1 = \frac{x^2}{2} + 2xy + 4xz$

$\Rightarrow \frac{\partial\phi_2}{\partial y} = f_2$

$\Rightarrow \frac{\partial\phi_2}{\partial y} = 2x-3y-z$

Integrating on bs

$\phi_2 = 2xy - \frac{3y^2}{2} - zy$

$\Rightarrow \frac{\partial\phi_3}{\partial z} = f_3$

$\Rightarrow \frac{\partial\phi_3}{\partial z} = 4x-y+2z$

Integrating on bs

$\phi_3 = 4xz - yz + z^2$

$\Rightarrow \phi = \phi_1 + \phi_2 + \phi_3$ (write common terms only once)

$\Rightarrow \boxed{\phi = \frac{x^2}{2} - \frac{3y^2}{2} + z^2 + 2xy - yz + 4xz + C}$

15. i) Prove that $\operatorname{div}(\operatorname{curl}\bar{F}) = \bar{0}$ ii) $\operatorname{curl}(\operatorname{grad}\phi) = \bar{0}$

i) If $\bar{F} = f_1\bar{i} + f_2\bar{j} + f_3\bar{k}$

$$\operatorname{curl}\bar{F} = \nabla \times \bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix} = \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}\right)\bar{i} - \left(\frac{\partial f_3}{\partial x} - \frac{\partial f_1}{\partial z}\right)\bar{j} + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y}\right)\bar{k}$$

$$\operatorname{div}(\operatorname{curl}\bar{F}) = \nabla \cdot (\nabla \times \bar{F})$$

$$= \left(\frac{\partial}{\partial x}\bar{i} + \frac{\partial}{\partial y}\bar{j} + \frac{\partial}{\partial z}\bar{k}\right) \cdot \left[\left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}\right)\bar{i} - \left(\frac{\partial f_3}{\partial x} - \frac{\partial f_1}{\partial z}\right)\bar{j} + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y}\right)\bar{k}\right]$$

$$= \frac{\partial^2 f_3}{\partial x \partial y} - \frac{\partial^2 f_2}{\partial x \partial z} - \frac{\partial^2 f_3}{\partial x \partial y} + \frac{\partial^2 f_1}{\partial y \partial z} + \frac{\partial^2 f_2}{\partial x \partial z} - \frac{\partial^2 f_1}{\partial y \partial z}$$

$\boxed{\operatorname{div}(\operatorname{curl}\bar{F}) = 0}$ Hence proved

ii) Consider $\nabla\phi = \frac{\partial\phi}{\partial x}\bar{i} + \frac{\partial\phi}{\partial y}\bar{j} + \frac{\partial\phi}{\partial z}\bar{k}$

$$\operatorname{curl}(\nabla\phi) = \nabla \times \nabla\phi$$

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial\phi}{\partial x} & \frac{\partial\phi}{\partial y} & \frac{\partial\phi}{\partial z} \end{vmatrix}$$

$$= \left[\frac{\partial}{\partial y}\left(\frac{\partial\phi}{\partial z}\right) - \frac{\partial}{\partial z}\left(\frac{\partial\phi}{\partial y}\right)\right]\bar{i} - \left[\frac{\partial}{\partial x}\left(\frac{\partial\phi}{\partial z}\right) - \frac{\partial}{\partial z}\left(\frac{\partial\phi}{\partial x}\right)\right]\bar{j} + \left[\frac{\partial}{\partial x}\left(\frac{\partial\phi}{\partial y}\right) - \frac{\partial}{\partial y}\left(\frac{\partial\phi}{\partial x}\right)\right]\bar{k}$$

$$= \left[\frac{\partial^2\phi}{\partial y \partial z} - \frac{\partial^2\phi}{\partial y \partial z}\right]\bar{i} - \left[\frac{\partial^2\phi}{\partial x \partial z} - \frac{\partial^2\phi}{\partial x \partial z}\right]\bar{j} + \left[\frac{\partial^2\phi}{\partial x \partial y} - \frac{\partial^2\phi}{\partial x \partial y}\right]\bar{k}$$

$\operatorname{curl}(\nabla\phi) = 0\bar{i} + 0\bar{j} + 0\bar{k}$

$\boxed{\operatorname{curl}(\nabla\phi) = \bar{0}}$ Hence proved

16. Show that $\text{div}(\text{grad } r^n) = \nabla^2(r^n) = n(n+1)r^{n-2}$ where $r = |\vec{r}|$

Sol: $\text{div}(\text{grad } r^n) = \nabla \cdot (\nabla r^n) = \nabla^2 r^n$ — (1)

$$= \frac{\partial^2 r^n}{\partial x^2} + \frac{\partial^2 r^n}{\partial y^2} + \frac{\partial^2 r^n}{\partial z^2} \quad \text{— (2)}$$

Consider $\frac{\partial^2 r^n}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}(r^n)\right]$

$$= \frac{\partial}{\partial x}\left[n r^{n-1} \frac{\partial r}{\partial x}\right] \qquad \frac{\partial r}{\partial x} = \frac{x}{r}$$

$$= \frac{\partial}{\partial x}\left(n r^{n-1} x \frac{x}{r}\right)$$

$$= \frac{\partial}{\partial x}(n r^{n-2} x)$$

$$= n r^{n-2}(1) + x\, n(n-2) r^{n-3} \frac{\partial r}{\partial x}$$

$$= n r^{n-2} + x\, n(n-2) r^{n-3} \frac{x}{r}$$

$$= n r^{n-2} + n(n-2) r^{n-4} x^2$$

Similarly $\frac{\partial^2 r^n}{\partial y^2} = n r^{n-2} + n(n-2) r^{n-4} y^2$ Substitute in (2)

$$\frac{\partial^2 r^n}{\partial z^2} = n r^{n-2} + n(n-2) r^{n-4} z^2$$

$$= n r^{n-2} + n r^{n-2} + n r^{n-2} + n(n-2) r^{n-4} x^2 + n(n-2) r^{n-4} y^2 + n(n-2) r^{n-4} z^2$$

$$= 3n r^{n-2} + n(n-2) r^{n-4}(x^2 + y^2 + z^2) \qquad (\because r^2 = x^2 + y^2 + z^2)$$

$$= 3n r^{n-2} + n(n-2) r^{n-4} r^2$$

$$= n r^{n-2}(3 + n - 2) = n(n+1) r^{n-2} \quad \text{— (3)} \qquad \text{From (1) \& (3)}$$

$\text{div}(\text{grad } r^n) = \nabla^2(r^n) = n(n+1) r^{n-2}$ Hence proved

17. Show that $\nabla^2[f(r)] = f''(r) + \frac{2}{r} f'(r)$

Sol. $\nabla^2[f(r)] = \frac{\partial^2}{\partial x^2}[f(r)] + \frac{\partial^2}{\partial y^2}[f(r)] + \frac{\partial^2}{\partial z^2}[f(r)]$ — ①

Consider $\frac{\partial^2}{\partial x^2}[f(r)] = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}[f(r)]\right]$

$= \frac{\partial}{\partial x}\left[f'(r)\frac{\partial r}{\partial x}\right]$ $\quad \frac{\partial r}{\partial x} = \frac{x}{r}$

$= \frac{\partial}{\partial x}\left[f'(r)\frac{x}{r}\right] \rightarrow (d(uvw))$

$= f'(r)x\left[\frac{\partial}{\partial x}\left(\frac{1}{r}\right)\right] + \frac{f'(r)}{r}\left(\frac{\partial x}{\partial x}\right) + \frac{x}{r} f''(r)\frac{\partial r}{\partial x}$

$= f'(r)x\left[\frac{-1}{r^2}\frac{\partial r}{\partial x}\right] + \frac{f'(r)}{r} + f''(r)\left(\frac{x^2}{r^2}\right)$

$= f'(r)x x x \frac{-1}{r^2} x \frac{x}{r} + \frac{f'(r)}{r} + f''(r)\left(\frac{x^2}{r^2}\right)$

$= f''(r)\frac{x^2}{r^2} + \frac{f'(r)}{r} - \frac{f'(r)x^2}{r^3}$

Similarly $\frac{\partial^2}{\partial y^2}[f(r)] = f''(r)\frac{y^2}{r^2} + \frac{f'(r)}{r} - f'(r)\frac{y^2}{r^3}$ Substitute in ①

$\frac{\partial^2}{\partial z^2}[f(r)] = f''(r)\frac{z^2}{r^2} + \frac{f'(r)}{r} - f'(r)\frac{z^2}{r^3}$

$\Rightarrow \nabla^2[f(r)] = \frac{f''(r)}{r^2}(x^2+y^2+z^2) + \frac{3f'(r)}{r} - \frac{f'(r)}{r^3}(x^2+y^2+z^2)$

$(\because r^2 = x^2+y^2+z^2)$

$= \frac{f''(r)}{r^2} x r^2 + \frac{3f'(r)}{r} - \frac{f'(r)}{r^3} x r^2$

$\Rightarrow \boxed{\nabla^2[f(r)] = f''(r) + \frac{2}{r} f'(r)}$ Hence proved